

Detailed Project Report

Topic:

- GUI Based Network Migration (Group 5)

Team Members:

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Technologies Used:

- Java
- Swing (Java GUI toolkit)
- SQLite (local database)
- JDBC (Java Database Connectivity)

As a group we built a Java GUI app that:

- Lets mobile IT workers input IP configuration settings (once per network)
- Stores these settings as reusable "profiles"
- Allows users to switch between saved profiles quickly (e.g., apply network settings when changing locations)
- Eliminates the need for users to manually track or re-enter network info

How the program works:

- User launches the program → the Main Menu appears. From the menu, the user can choose to:
 - Add a new profile: input IP address, subnet, gateway, DNS, and save it.
 - Switch to a profile: select from saved profiles and simulate applying it.
- When finished, the user can return to the Main Menu or exit.

Key System Features:

- Add Profile: GUI form to input IP address, subnet mask, gateway, DNS
- Save Profile: Information is stored in a local SQLite DB
- Switch Profile: Dropdown to select a profile and view network settings
- Simulated Migration: Displays selected IP configuration (simulating switch)

Java Classes:

- Main.java: The app's entry point. Ensures the DB table exists and launches MainMenu.
- MainMenu.java: The central navigation screen. Allows users to select Add or Switch.
- AddProfileForm.java: A form where the user can enter network settings and save them as a profile.

- SwitchProfileForm.java: Lets the user select a previously saved profile and switch to it.

Data Management:

- We used SQLite where the profiles were stored in a table called “profiles”
- Each profile includes: network_name, ip_address, subnet_mask, gateway, dns

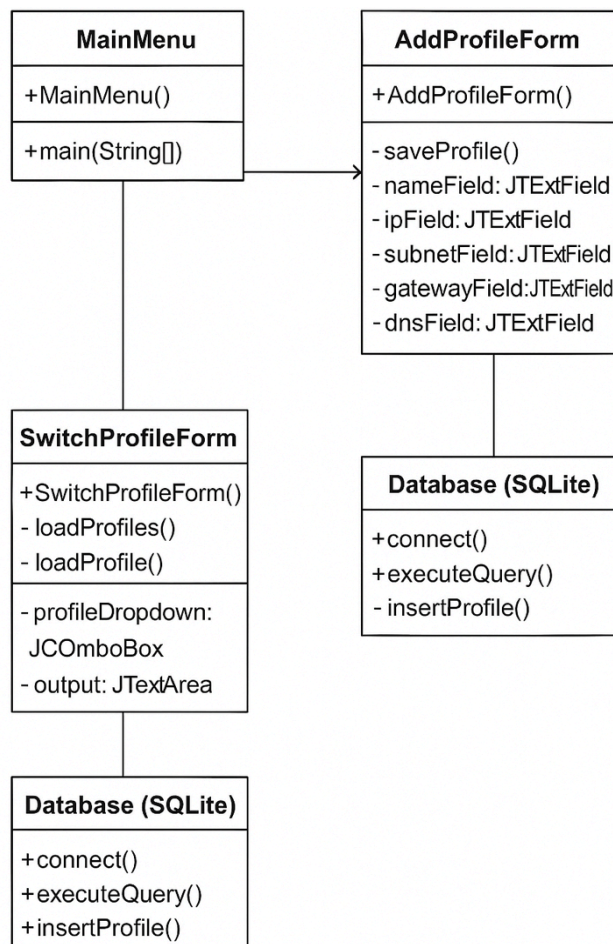
How to run program:

- Open terminal and navigate to the src/ folder.
- Compile the project with:
 - javac -cp ".../lib/sqlite-jdbc-3.49.1.0.jar" *.java
- Run the project with:
 - java -cp ".../lib/sqlite-jdbc-3.49.1.0.jar" Main
- The main menu should load.

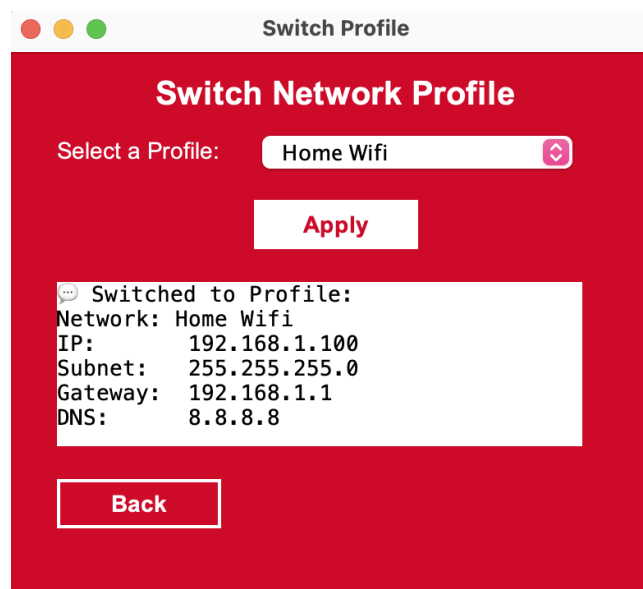
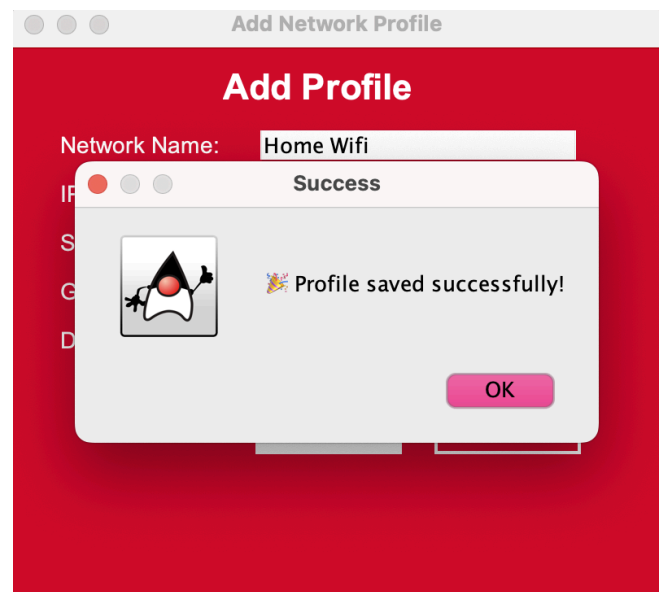
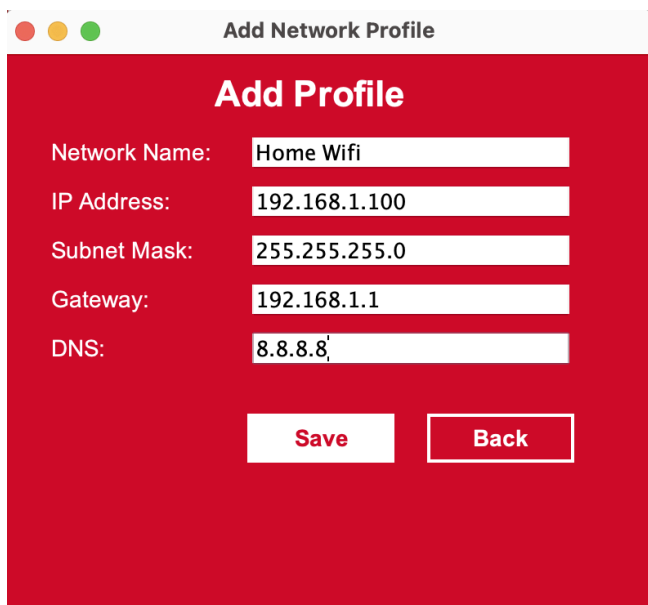
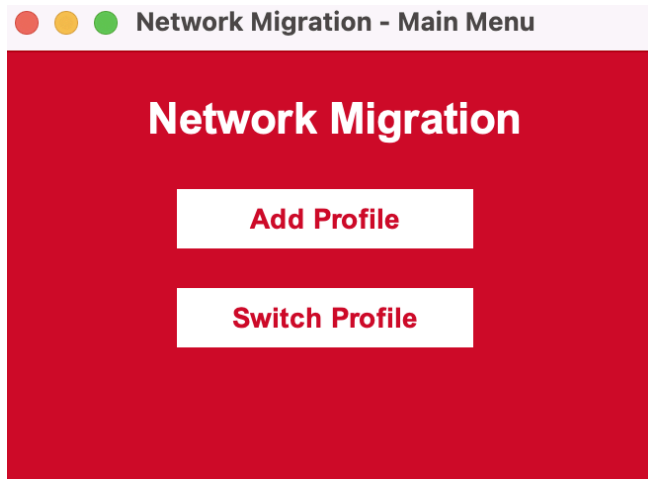
System Architecture:

- MainMenu.java → opens → AddProfileForm.java / SwitchProfileForm.java
- SQLite → stores and loads → profiles.db

UML Diagram:



Screenshots:



Dropdown Menu would show more profiles the worker would be able to switch to. Thus reducing the need to reenter information numerous times.

Conclusion:

- This application simplifies network transitions for IT workers by storing and retrieving complete IP configurations with one-time input. It reduces the need for manual re-entry and supports quick profile switching.